Kyung Hyun Lee

AI Researcher in Physiological and Clinical Time-Series Learning

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Summary

AI researcher developing and evaluating machine learning models for physiological signals and longitudinal clinical data, with particular attention to external validation and continuous clinical monitoring. PhD student in Digital Health at SAIHST, Sungkyunkwan University, and AI researcher at AITRICS. Co-founder & CEO of BreathYou Co., Ltd.

Education

PhD Student — **Sungkyunkwan University (SAIHST)**, Digital Health — Seoul, South Korea — 2026 – present
- Advisor: Prof. Byung-Jae Lee
- Research on Transformer-based trajectory models of respiratory health

Master of Science — **Sungkyunkwan University (SAIHST)**, Digital Health — Seoul, South Korea — 2019 – 2021
- Advisor: Prof. Mira Kang
- M Lab. — Statistical analysis and predictive model development using medical big data

Bachelor of Science — **Incheon National University**, Electronic Engineering & Computational Neuroscience — Incheon, South Korea — 2014 – 2019
- Advisor: Prof. Sangwon Byun
- Bioelectronics Lab. — Biosignal (ECG, EMG) data collection, signal analysis, and real-time prediction model development

Experience

University of Seoul, Lecturer — Seoul, South Korea — 2026 – present — 1 year

Faculty appointment in the Department of Sport Science, teaching the undergraduate course Exercise and Digital Healthcare.
- Designed and teach Exercise and Digital Healthcare, a fourth-year major elective (Fall 2026)

BreathYou Co., Ltd., Co-founder & CEO — South Korea — 2025 – present — 1 year

Digital health startup focused on allergy and respiratory AI, developing DeepBreath CDSS and MediPipe.
- Selected for 2025 Pre-startup Package
- Incorporated on 5 August 2026
- DeepBreath CDSS: Clinical decision support for predicting pulmonary function decline trajectories
- MediPipe: OCR/NLP pipeline for medical record processing

AITRICS, Researcher (2024 – present) · Industrial Technical Personnel — Alternative Military Service (2022 – 2024) — Seoul, South Korea — 2022 – present — 4 years

Medical data (RWD) cleaning, analysis pipeline construction, predictive model development, and external validation. Contributed to the cardiac arrest early-warning model that obtained MFDS (Korean FDA) approval as an AI medical device, and currently leading product development of a self-developed AKI prediction model toward MFDS approval.

- Research design for clinical data analysis projects
- Data extraction and cleaning from multi-center hospitals
- Automated data analysis pipelines using Airflow
- Deep learning-based clinical time-series prediction models
- Clinical simulation-based model evaluation and survival analysis
- Cardiac arrest prediction model — supported MFDS approval as an AI medical device
- AKI prediction model — developed the model and currently leading productization toward MFDS approval

Exosystems, Biosignal AI Research Engineer — South Korea — May 2022 – Aug 2022 — 4 months

Deep learning-based EMG modeling for Muscle Quality monitoring.

- RNN, CRNN-based Muscle Quality time-series prediction model development
- Clinical validation-oriented AI model evaluation

Incheon National University, Researcher — Incheon, South Korea — Sept 2021 – May 2022 — 9 months

- EMG-based real-time hand gesture recognition algorithm R&D
- Deep learning-based ECG modeling for autonomic nervous system monitoring

Samsung Medical Center & SAIHST, Researcher — Seoul, South Korea — Dec 2018 – Aug 2021 — 2 years 9 months

EHR data extraction and analysis from CDW; development of ML-based CDSS.

- Data cleaning and health checkup database construction
- Text mining of unstructured medical records
- Statistical analysis and predictive/diagnostic model development

Teaching

University of Seoul, Department of Sport Science: Lecturer · Exercise and Digital Healthcare, fourth-year undergraduate elective (Fall 2026)

University of Seoul, Department of Sport Science: Invited seminar · Healthcare industry, two hours, hosted by Prof. Sae Young Jae (2024)

Incheon National University: Guest lecture · Student-Designed Convergence Major, one hour (2024)

Awards

Pre-TeX-Corps Business Plan Presentation — Excellence Award — 2026

Sungkyunkwan University, Laboratory Startup Innovation Center

- Business plan presentation for the Ministry of Science and ICT TeX-Corps laboratory startup exploration program
- MediPipe: on-premise medical data infrastructure software that turns in-hospital test results and clinical records into AI-ready data

Lab Startup Competition — Excellence Award — 2026

Sungkyunkwan University, Start-up Support Foundation

- University-wide competition among Sungkyunkwan University graduate-student laboratory startup clubs
- MediPipe: on-premise medical data infrastructure software that turns in-hospital test results and clinical records into AI-ready data

Consortium Lab-based Startup Camp — Excellence Award — 2026

Sungkyunkwan University, Startup-Centered University program

- Consortium competition among laboratory startup teams from universities across the Seoul metropolitan area
- MediPipe: the data infrastructure that lets hospitals use AI

Human Understanding AI Paper Competition — Top 7 & ETRI Director's Award 2022
ETRI, Ministry of Science and ICT
- Multimodal Attention-based Korean emotion recognition model combining RoBERTa (Text) and Wav2vec 2.0 (Audio)

OCR AI Training Data Hackathon — Excellence Award 2021
Ministry of Science and ICT, NIA
- AI-Assisted OCR Annotation System for pharmaceutical products (CRAFT-based)

Medical Information Analysis Expert Advanced Course — Grand Prize 2021
Ministry of Health and Welfare, Asan Medical Center
- Real-time BIS prediction model using intraoperative EEG data (25-second lookahead)

Medical AI Expert Training Course — Excellence Award 2020
Korea Human Resource Development Institute for Health & Welfare
- 30-day mortality and ICU admission prediction model using preoperative and intraoperative vital signs

STAR Program Paper Presentation — Grand Prize 2018
DSP-IN-U Smart Device Talent Development Program
- EMG-based real-time hand gesture prediction model development

Academic Service

Journal Reviewer: Computers in Biology and Medicine (4), Frontiers in Medicine (2), Frontiers in Digital Health, Frontiers in Network Physiology, Frontiers in Psychiatry, Information Processing & Management · 2026–present

Conference Reviewer: NeurIPS Workshop: Responsible Communication of Machine Learning Research in Biomedicine (RCMLR) · 2026

Certificates

Medical Information Analysis Expert Advanced Course Aug 2021
Korea Human Resource Development Institute for Health & Welfare, Asan Medical Center

Google Developer Machine Learning Boot Camp Jan 2021
Google Developers

Medical AI Expert Training Course Dec 2020
Korea Human Resource Development Institute for Health & Welfare

Skills

Programming & ML: Python, PyTorch, TensorFlow, Scikit-learn

Data & Database: SQL (PostgreSQL), Airflow, Data Pipeline

Statistics & Analysis: R, Survival Analysis, Clinical Statistics

Web Development: FastAPI, Flask, JavaScript, HTML/CSS, AWS

Languages

Korean
Native speaker

English
Professional working proficiency

References

Prof. Sangwon Byun
Incheon National University, Department of Electronic Engineering; Bachelor's advisor
(Bioelectronics Lab. · Scholar)

Prof. Mira Kang
Samsung Medical Center; Master's advisor at SAIHST, Sungkyunkwan University

Prof. Jin-Young Lee
Samsung Medical Center, Health Promotion Center

Prof. Byung-Jae Lee
Samsung Medical Center, Department of Allergy and Clinical Immunology; Doctoral advisor at SAIHST, Sungkyunkwan University

Prof. Sae Young Jae
University of Seoul, Department of Sport Science; research collaborator